

COMPUTER ARCHITECTURE



Hamid Sarbazi-Azad

Department of Computer Engineering
Sharif University of Technology
Tehran, Iran

What is computer architecture?

Computer architecture, like other architecture fields, is the **art** of determining the **needs** of the user of a **structure**¹ and then designing it to **meet** those needs as effectively as possible within economical/technological **constraints**.

Computer architecture is a set of **rules and methods** that describe the **functionality**, **organization**², and **implementation**³ of computer systems.

¹ **Structure**: static arrangement of the parts

² **Organization**: dynamic interaction of parts and their control

³ **Implementation**: design of specific building blocks

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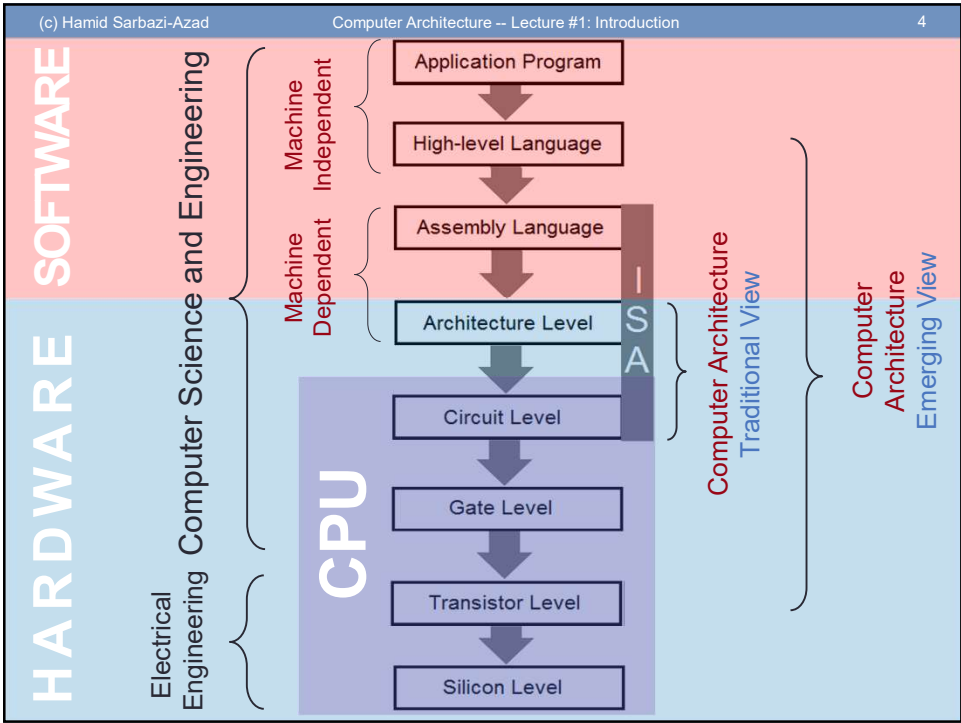
What is computer architecture?

CA involves designing instruction set architecture, microarchitecture and logic, and implementing computer systems.

It works with laws & walls

Questions:

- Where does CA stand in Computer Science and Engineering?
- Where is Hardware?
- Where is Software?
- What about Electrical Engineering (AKA digital electronic systems in Iran)?



Why computer architecture (CA)?

Because CA is perhaps the most fundamental subject in computer science. Without computers, the field of computer science does not exist.

Whatever we do, be it surfing the web, sending email, or typing a document, is on top of computer architecture.

Computer architecture shows us how machines are designed, built, and operate.

Knowing what is inside, and how it works will help you design, develop, and implement applications/programs better, faster, cheaper, more efficient, and easier to use.

High-performance CA

High-performance computer architecture (HPCA) is the art of designing and building efficient computer systems for massive computation. HPCA techniques and advances in integration technology (thanks to Moore's law) have fuelled the thirst for building very powerful computers (supercomputers) during the last six decades.

Supercomputers are evaluated every 6 months and reported in www.top500.org based on their computing power in FLOPS (Floating-point Operations Per Second).

$$1 \text{ EFLOPS} = 10^3 \text{ PFLOPS} = 10^6 \text{ TFLOPS} = 10^9 \text{ GFLOPS} \\ = 10^{12} \text{ MFLOPS} = 10^{15} \text{ KFLOPS} = 10^{18} \text{ FLOPS}$$

Supercomputers (some successful systems)

Cray I

Year 1975

- 160 MFLOPS
- Vector machine
- 8 MB main memory



Supercomputers (some successful systems)

Cray X-MP

Year 1982

- 1 GFLOPS
- Shared-memory multiprocessor with 2 to 4 enhanced Cray-I based vector processors



Supercomputers (some successful systems)

The Earth Simulator

- Operational in late 2002
- Result of 5-year design and implementation effort
- Equivalent power to top 15 US supercomputers @ 2002

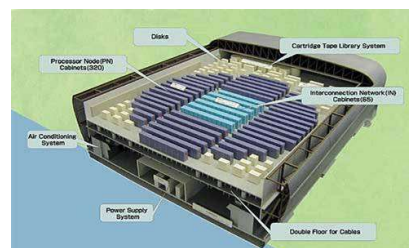
It had:

- 640 nodes
- 8 vector processors per node, 5120 total
- 8 GFLOPS per processor, 40 TFLOPS total
- 16 GB memory per node, 10 TB total
- 2800 km of cables among 320 cabinets (2 nodes each)
- Cost: US\$350 M

Supercomputers (some successful systems)



The Earth Simulator



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Supercomputers (some successful systems)

IBM BlueGene

Massively distributed shared-memory multicomputer

Delivered: 2004-2007

It had:

Peak performance: 360 TFLOPS

Number of nodes: 65536

Node: 2 processors each having a double floating-point unit

Cost: US\$130 M



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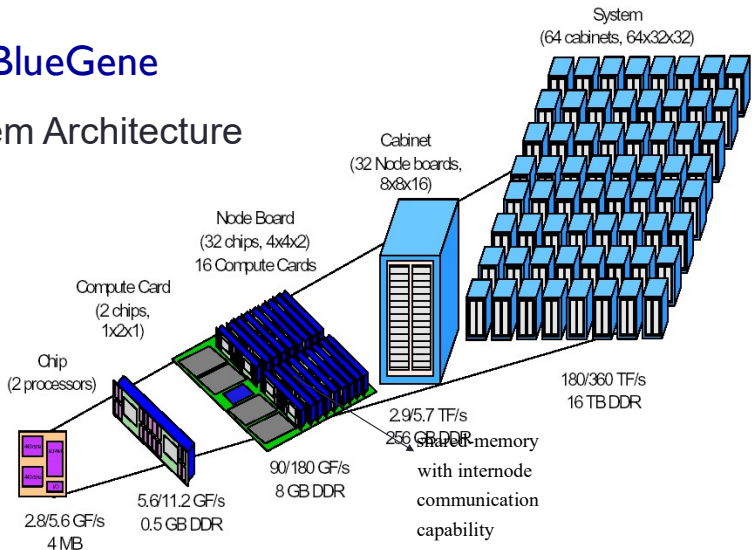
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Supercomputers (some successful systems)

IBM BlueGene

System Architecture



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Supercomputers (some successful systems)

Cray X1

NUMA Scalable up to 1024 Nodes

The diagram illustrates the Cray X1 architecture. It features a central, light-blue, cloud-like shape labeled "Interconnection Network". Surrounding this central network are several green rectangular blocks, each representing a node. Each node contains a grid of smaller colored squares (blue, red, green, yellow) representing processors and memory. Bidirectional arrows connect each node to the central network. Below the network, there are three dots and the text "inter - cluster message passing mechanism".

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Supercomputers (some successful systems)

Cray X1

Cray Supercomputer Evolution and Roadmap

The diagram shows the evolution of Cray supercomputers as a line graph. The x-axis represents time, and the y-axis represents performance. The line starts at "Cray-1" and goes through "X-MP", "Y-MP", "C90", "T90", "J90", "SV1", and "SV1ex". A callout box for "Microprocessor-based MPP" (1000's of processors, UNICOS/mk) points to the "C90" node. Another callout box for "High Bandwidth Vector" (1-32 processors) points to the "SV1ex" node. The line then branches into two paths: one leading to "T3D", "T3E", "1200", and "1350", and another leading to "X1", "X1e", and "BlackWidow". A callout box for "High BW MPP" points to the "X1" node. The line continues to "BW+", "BW++", and "DARPA HPCS". A yellow callout box for "sustained petaflops" points to the "DARPA HPCS" node. A red callout box for "by 2010-2015 time frame" points to the "DARPA HPCS" node.

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Supercomputers (some successful systems)

IBM Roadrunner

Built in 2008, as the world’s first PFLOPS machine, it uses Cell (multi-core) processors designed by Sony, Toshiba and IBM, and introduced the new line of Low-Cost Supercomputing.

Total cost: US\$133 M
Processors: 12960 Cell processors (each of 9-cores)
Peak performance: 1.33 PFLOPS (1.33x10¹⁵ FLOPS)
Power: 3.9 MW
Area: 296 Racks, 511 m²
Weight: 227 Tons

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Supercomputers (Top 10)

Nov. 2025 List

Rank	System	Cores	Rmax (PFlops)	Rpeak (PFlops)	Power (kW)
1	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS United States	11,340,000	1,809.00	2,821.10	29,685
2	Frontier - HPE Cray EX235a, AMD Optimized Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, Cray OS United States	9,066,176	1,353.00	2,055.72	24,607
3	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Slingshot-11 United States	9,264,128	1,012.00	1,980.01	38,698
4	JUPITER Booster - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, InfiniBand NDR200, RedHat Enterprise Linux Germany	4,801,344	1,000.00	1,226.28	15,794
5	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA InfiniBand NDR, Microsoft Azure United States	2,073,600	561.20	846.84	-
6	HPC6 - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, RHEL 8.9 Italy	3,143,520	477.90	606.97	8,461
7	Fugaku - A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
8	Alps - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS, HPE Swiss National SC Switzerland	2,121,600	434.90	574.84	7,124
9	LUMI - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	531.51	7,107
10	Leonardo - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, InfiniBand, EVIDEN EuroHPC/CINECA Italy	1,824,768	241.20	306.31	7,494

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Supercomputers (some successful systems)

Frontier

Total cost: ~ US\$ 600M

Core#: ~ 9 M cores

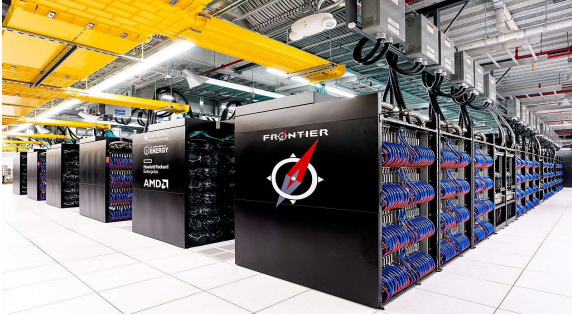
Peak performance: ~ 2 EFLOPS

Max performance: ~ 1.36 EFLOPS

Memory: ~ 5 Peta Bytes

Power: 24.6 MW

Years built: 2021-2022



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Frontier v.s. human brain?

In 2014, Japanese researchers used K-Computer (4th fastest supercomputer in the world at the time) to crunch the calculations for a single second of part of brain activity in 40 minutes! (see <https://www.scienceabc.com/humans/the-human-brain-vs-supercomputers-which-one-wins.html>)

European Union has started Human Brain Project in 2013, to fully simulate human brain (see <https://www.humanbrainproject.eu/en/>).

Frontier

~ 1.36 EFLOPS 😊

~ 1400 m³ 😞

~ 24 MW 😞

~ US\$ 600 M 😊

Compare to US\$ 446 M for A380



Human Brain

1.2 PFLOPS¹ 😊

1400 cm³ (~0.001m³) 😊

20 W 😊

? 😞

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Topics covered:

1. Overview of
 - Logic design,
 - Data representation
 - Machine language and structure
2. Memory system
 - Memory hierarchy and locality philosophy
 - Memory technology
 - Cache memory
 - Virtual memory
3. Arithmetic unit
 - Add/Sub/Multiply/Divide schemes for fixed-point binary numbers
 - Add/Sub/Multiply/Divide schemes for fixed-point decimal numbers
 - Add/Sub/Multiply/Divide schemes for floating-point numbers
 - Special-purpose arithmetic circuits

Topics covered (cont.):

4. Control Unit
 - Hardwired control
 - Microprogrammed control
5. RISC processors
6. Pipelining
7. Input/Output (I/O) organization
8. Introduction to Advanced Computer Organization

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References

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5. V. Heuring, H. Jordan, **Computer systems design & architecture**, Prentice-Hall.
6. J. Cavanagh, **Digital computer arithmetic, design & implementation**, McGraw-Hill.
7. K. Hwang, **Computer arithmetic: principles, architecture & design**, Wiley.

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Marking Scheme

Midterm Exam	20%	*** 24 th Ordibehesht 1405 ***
Final Exam	30%	
Project	20%	
Homework	30%	

Computer Architecture team



Lecturer: Hamid Sarbazi-Azad
Email: azad@sharif.edu / azad@ipm.ir
Phone: 66166622
Room: 621



Head TA: Atiyeh Gheibi-Fetrat
Email: atiye.gheibi@gmail.com
Phone: 66166672
Room: HPCAN Lab. (703)

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